4.1 Interpreting Derivatives in Context — teacher key

Question: What does a derivative tell you when the function is a real quantity, like water in a tank?

Self-paced lesson

No clock, no trimming, no skipping, no end mark, no stamp. Every tab is required in order; progress saves on the device so a student can stop and finish the next day.

Warm-up review (3 easy items)

- The derivative of $5x^2 - 3x$ is... **10x − 3.** Power rule on each term: $5x^2$ gives $10x$, and $-3x$ gives -3.
- A line passes through $(1, 2)$ and $(3, 8)$. Its slope is... **3.** Rise over run: $\dfrac{8-2}{3-1} = 3$.
- If $f'(x) < 0$ on an interval, then f is... **decreasing.** A negative slope means the graph of f goes down.

Explore stations

Four hands-on stations (X1, X2, X3, X4). Station answers are in the teacher view of the lesson page (add `#key` to its address).

Explore: what did you see

- In Station 1, at the moment the tank was neither filling nor emptying, the rate reading was... **0.** A flat tangent has slope 0, and the slope is the rate.
- In Station 3, when the step h was small, the tangent-line estimate was... **very close to the actual value.** For a small step the curve stays close to its tangent line, so the estimate is good.

Mid-lesson check (Huddle tab)

- $V'(2) = 8$ gallons per minute. About how many gallons come in during the next half minute? **4.** Change is about rate times time: $8 \times 0.5 = 4$ gallons.
- If $C'(a) < 0$, then at $x = a$ the cost is... **going down.** A negative rate means the quantity is falling at that point.

Targets

- I can say what f?(a) means in a story, with units.
- I can tell if a quantity is rising or falling from the sign of f?.
- I can use f(a) and f?(a) to estimate a nearby value.

Predictions

- A tank holds $V(t)$ gallons after t minutes, and $V'(5) = -3$. At $t = 5$ the tank is... **losing water.** A negative derivative means the amount is going down.
- For the same tank, $V(t)$ is in gallons and t is in minutes. The derivative $V'(t)$ is measured in... **gallons per minute.** A rate is the amount that changes divided by the time it takes: gallons per minute.

Huddle script (answers in green)

Definitions

- **Derivative as a rate:** f?(a) is the **rate** of change of f at x = a. It is the **slope** of the tangent line.
- **Units:** The units of f? are the units of f **per** the units of x.
- **Reading the sign:** If f?(a) > 0, then f is **increasing**. If f?(a) < 0, then f is **decreasing**.
- **Estimating:** f(a + h) ≈ f(a) + **h** · f?(a). It works best when h is **small**.

Worked example

A tank holds V(t) = 40 + 12t – t^2 gallons, t in minutes. Find V?(3) and say what it means.

- Differentiate: V?(t) = 12 – **2t**
- Put in t = 3: V?(3) = 12 – 6 = **6**
- The units are gallons **per minute**
- At t = 3 minutes, the tank is **filling** at 6 gallons per minute.

Label the picture

1. The curve: its height is the **amount**.
2. The tangent line: its slope is the **rate**.
3. The horizontal axis: the **time** or input.

Every level, with answers

Say what it means

Level 1. $T(t)$ is the temperature, in degrees Fahrenheit, of an oven t minutes after it is turned on. Also $T'(7) = 8$. What does this say?
(A) The temperature is 8 degrees when t = 7. **(B) When t = 7, the temperature is rising at 8 degrees per minute.** (C) When t = 7, the temperature is rising at 8 minutes per degree. (D) When t = 7, the temperature is falling at 8 degrees per minute.
Answer: B The sign is positive, so the temperature is rising. The size is 8 and the units are degrees per minute.

Level 2. $C(x)$ is the cost, in dollars, of making x shirts. Also $C'(80) = -8$. What does this say?
(A) When x = 80, the cost is rising at 8 dollars per shirt. (B) The cost is 8 dollars when x = 80. **(C) When x = 80, the cost is falling at 8 dollars per shirt.** (D) When x = 80, the cost is falling at 8 shirts per dollar.
Answer: C The sign is negative, so the cost is falling. The size is 8 and the units are dollars per shirt.

Level 3. $H(t)$ is the height, in feet, of a tree t years after it was planted. Which equation says: "When t = 1, the height is growing at 7 feet per year"?
(A) $H(1) = 7$ **(B)** $H'(1) = 7$ (C) $H'(7) = 1$ (D) $H(7) = 1$
Answer: B A rate is a derivative, so use H'. The input is 1 and the rate is 7: $H'(1) = 7$.

Level 4. $H(t)$ is the height, in feet, of a tree t years after it was planted. Which equation says: "When t = 11, the height is growing at 3 feet per year"?
(A) $H(11) = 3$ **(B) $H'(11) = 3$** (C) $H(3) = 11$ (D) $H'(3) = 11$
Answer: B A rate is a derivative, so use H'. The input is 11 and the rate is 3: $H'(11) = 3$.

Units of a rate

Level 1. $T(t)$ is the temperature, in degrees Fahrenheit, of an oven t minutes after it is turned on. What are the units of $T'(t)$?
(A) degrees per minute (B) degrees minutes (C) degrees (D) minutes per degrees
Answer: A The output is in degrees and the input is in minutes, so the derivative is in degrees per minute.

Level 2. $C(x)$ is the cost, in dollars, of making x shirts. What are the units of $C''(x)$?
(A) dollars per shirt **(B) dollars per shirt per shirt** (C) dollars per shirt squared times shirt (D) dollars
Answer: B The output is in dollars and the input is in shirts, so the second derivative is in dollars per shirt per shirt.

Level 3. $W(h)$ is the volume, in cubic feet, of water in a rain barrel after h hours of rain. What are the units of $W'(h)$?
(A) hours per cubic feet (B) cubic feet hours (C) cubic feet **(D) cubic feet per hour**
Answer: D The output is in cubic feet and the input is in hours, so the derivative is in cubic feet per hour.

Level 4. $W(h)$ is the volume, in cubic feet, of water in a rain barrel after h hours of rain. What are the units of $W'(h)$?
(A) cubic feet (B) hours per cubic feet **(C) cubic feet per hour** (D) cubic feet hours
Answer: C The output is in cubic feet and the input is in hours, so the derivative is in cubic feet per hour.

Estimate a nearby value

Level 1. $T(t)$ is the temperature, in degrees Fahrenheit, of an oven t minutes after it is turned on. Right now $t = 10$, with $T(10) = 330$ and $T'(10) = 8$ degrees per minute. Estimate $T(13)$.
Answer: 354 degrees The step is 3 minutes. Change $\approx 8 \times 3 = 24$. So $T(13) \approx 330 + 24 = 354$.

Level 2. $C(x)$ is the cost, in dollars, of making x shirts. Right now $x = 83$, with $C(83) = 790$ and $C'(83) = -2$ dollars per shirt. Estimate $C(80)$.
Answer: 796 dollars The step is -3 shirts. Change $\approx$ -2 $\times$ -3 = 6. So $C(80) \approx 790 + 6 = 796$.

Level 3. $H(t) = -t^3 - 3t^2 + 13t + 12$ is the height, in feet, of a tree t years after it was planted. Use $H(1)$ and $H'(1)$ to estimate $H(4)$.
Answer: 33 feet $H(1) = 21$ and $H'(1) = 4$. The estimate is $21 + 3 \cdot (4) = 33$.

Level 4. $H(t) = -3t^2 + 10t + 50$ is the height, in feet, of a tree t years after it was planted. Use $H(6)$ and $H'(6)$ to estimate $H(7)$.
Answer: -24 feet $H(6) = 2$ and $H'(6) = -26$. The estimate is $2 + 1 \cdot (-26) = -24$.

Find the rate

Level 1. $T(t) = -2t^2 + 13t + 48$ is the temperature, in degrees Fahrenheit, of an oven t minutes after it is turned on. Find $T'(3)$.
Answer: 1 degrees per minute $T'(t) = -4t + 13$. Putting in $t = 3$ gives 1 degrees per minute.

Level 2. $C(x) = 2\sqrt{x}$ is the cost, in dollars, of making x shirts. Find $C'(16)$.
Answer: 0.25 dollars per shirt $C'(x) = \dfrac{2}{2\sqrt{x}}$. Putting in $x = 16$ gives 0.25 dollars per shirt.

Level 3. $H(t) = 70e^{0.2t}$ is the height, in feet, of a tree t years after it was planted. Find $H'(0)$.
Answer: 14 feet per year $H'(t) = 70 \cdot 0.2e^{0.2t}$. Putting in $t = 0$ gives 14 feet per year.

Level 4. $H(t) = (t^2 + 1)^3$ is the height, in feet, of a tree t years after it was planted. Find $H'(1)$.
Answer: 24 feet per year $H'(t) = 3(t^2 + 1)^2 \cdot 2t$. Putting in $t = 1$ gives 24 feet per year.

Common misconceptions

Students say	Say instead
V?(3) = 6 means the tank holds 6 gallons at 3 minutes.	V(3) is the amount. V?(3) is the rate: 6 gallons per minute.
The units of V? are gallons.	A rate has units of the output per units of the input: gallons per minute.
A negative derivative means the quantity is negative.	It means the quantity is going down. The quantity itself can still be large.
The estimate f(a) + h f?(a) is exact.	It is an estimate. It is best for small h and gets worse as h grows.
Forgetting the units or the direction when writing what f? means.	Write rising or falling, then the size, then "units per unit".